

OpenStax Calculus Volume 3

Section 2.7 Exercises: Cylindrical and Spherical Coordinates

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Use the following figure as an aid in identifying the relationship between the rectangular, cylindrical, and spherical coordinate systems.

[Figure: This figure is the first octant of the 3-dimensional coordinate system. There is a line segment from the origin upwards. It is labeled “rho.” The angle between this line segment and the z-axis is labeled “phi.” There is also a broken line from the origin to the shadow of the point. This line segment is in the xy-plane and is labeled “r.” The angle between r and the x-axis is labeled “theta.”]

For the following exercises, the cylindrical coordinates (r, θ, z) of a point are given. Find the rectangular coordinates (x, y, z) of the point.

1. $(4, \frac{\pi}{6}, 3)$
2. $(3, \frac{\pi}{3}, 5)$
3. $(4, \frac{7\pi}{6}, 3)$
4. $(2, \pi, -4)$

For the following exercises, the rectangular coordinates (x, y, z) of a point are given. Find the cylindrical coordinates (r, θ, z) of the point.

5. $(1, \sqrt{3}, 2)$
6. $(1, 1, 5)$
7. $(3, -3, 7)$
8. $(-2\sqrt{2}, 2\sqrt{2}, 4)$

For the following exercises, the equation of a surface in cylindrical coordinates is given.

Find an equation of the surface in rectangular coordinates. Identify and graph the surface.

9. [T] $r = 4$
10. [T] $z = r^2 \cos^2 \theta$
11. [T] $r^2 \cos(2\theta) + z^2 + 1 = 0$
12. [T] $r = 3 \sin \theta$

13. [T] $r = 2 \cos \theta$
14. [T] $r^2 + z^2 = 5$
15. [T] $r = 2 \sec \theta$
16. [T] $r = 3 \csc \theta$

For the following exercises, the equation of a surface in rectangular coordinates is given. Find an equation of the surface in cylindrical coordinates.

17. $z = 3$
18. $x = 6$
19. $x^2 + y^2 + z^2 = 9$
20. $y = 2x^2$
21. $x^2 + y^2 - 16x = 0$
22. $x^2 + y^2 - 3\sqrt{x^2 + y^2} + 2 = 0$

For the following exercises, the spherical coordinates (ρ, θ, ϕ) of a point are given. Find the rectangular coordinates (x, y, z) of the point.

23. $(3, 0, \pi)$
24. $(1, \frac{\pi}{6}, \frac{\pi}{6})$
25. $(12, -\frac{\pi}{4}, \frac{\pi}{4})$
26. $(3, \frac{\pi}{4}, \frac{\pi}{6})$

For the following exercises, the rectangular coordinates (x, y, z) of a point are given. Find the spherical coordinates (ρ, θ, ϕ) of the point. Express the measure of the angles in degrees rounded to the nearest integer.

27. $(4, 0, 0)$
28. $(-1, 2, 1)$
29. $(0, 3, 0)$
30. $(-2, 2\sqrt{3}, 4)$

For the following exercises, the equation of a surface in spherical coordinates is given. Find an equation of the surface in rectangular coordinates. Identify and graph the surface.

31. [T] $\rho = 3$
32. [T] $\phi = \frac{\pi}{3}$
33. [T] $\rho = 2 \cos \phi$
34. [T] $\rho = 4 \csc \phi$
35. [T] $\phi = \frac{\pi}{2}$

36. [T] $\rho = 6 \csc \phi \sec \theta$

For the following exercises, the equation of a surface in rectangular coordinates is given. Find an equation of the surface in spherical coordinates. Identify the surface.

37. $x^2 + y^2 - 3z^2 = 0, z \neq 0$

38. $x^2 + y^2 + z^2 - 4z = 0$

39. $z = 6$

40. $x^2 + y^2 = 9$

For the following exercises, the cylindrical coordinates of a point are given. Find its associated spherical coordinates, with the measure of the angle ϕ in radians rounded to four decimal places.

41. [T] $(1, \frac{\pi}{4}, 3)$

42. [T] $(5, \pi, 12)$

43. $(3, \frac{\pi}{2}, 3)$

44. $(3, -\frac{\pi}{6}, 3)$

For the following exercises, the spherical coordinates of a point are given. Find its associated cylindrical coordinates.

45. $(2, -\frac{\pi}{4}, \frac{\pi}{2})$

46. $(4, \frac{\pi}{4}, \frac{\pi}{6})$

47. $(8, \frac{\pi}{3}, \frac{\pi}{2})$

48. $(9, -\frac{\pi}{6}, \frac{\pi}{3})$

For the following exercises, find the most suitable system of coordinates to describe the solids.

49. The solid situated in the first octant with a vertex at the origin and enclosed by a cube of edge length a , where $a > 0$

50. A spherical shell determined by the region between two concentric spheres centered at the origin, of radii of a and b , respectively, where $b > a > 0$

51. A solid inside sphere $x^2 + y^2 + z^2 = 9$ and outside cylinder $(x - \frac{3}{2})^2 + y^2 = \frac{9}{4}$

52. A cylindrical shell of height 10 determined by the region between two cylinders with the same center, parallel rulings, and radii of 2 and 5, respectively

53. [T] Use a CAS to graph the region between elliptic paraboloid $z = x^2 + y^2$ and cone $x^2 + y^2 - z^2 = 0$. Then describe the region in cylindrical coordinates.

54. [T] Use a CAS to graph in spherical coordinates the “ice cream-cone region” situated above the xy -plane between sphere $x^2 + y^2 + z^2 = 4$ and elliptical cone $x^2 + y^2 - z^2 = 0$.

55. Washington, DC, is located at 39°N and 77°W (see the following figure). Assume the radius of Earth is 4000 mi. Express the location of Washington, DC, in spherical coordinates.

[Figure: This figure is an image of a globe. On the globe there is a point labeled where

Washington, DC is located. It is labeled with 39 degrees north latitude and 77 degrees west longitude./

56. San Francisco is located at 37.78°N and 122.42°W . Assume the radius of Earth is 4000 mi. Express the location of San Francisco in spherical coordinates.
57. Find the latitude and longitude of Rio de Janeiro if its spherical coordinates are $(4000, -43.17^\circ, 102.91^\circ)$.
58. Find the latitude and longitude of Berlin if its spherical coordinates are $(4000, 13.38^\circ, 37.48^\circ)$.
59. [T] Consider the torus of equation $(x^2 + y^2 + z^2 + R^2 - r^2)^2 = 4R^2(x^2 + y^2)$, where $R \geq r > 0$.
 - (a) Write the equation of the torus in spherical coordinates.
 - (b) If $R = r$, the surface is called a horn torus. Show that the equation of a horn torus in spherical coordinates is $\rho = 2R \sin \phi$.
 - (c) Use a CAS to graph the horn torus with $R = r = 2$ in spherical coordinates.
60. [T] The “bumpy sphere” with an equation in spherical coordinates is $\rho = a + b \cos(m\theta) \sin(n\phi)$, with $\theta \in [0, 2\pi]$ and $\phi \in [0, \pi]$, where a and b are positive numbers and m and n are positive integers, may be used in applied mathematics to model tumor growth.
 - (a) Show that the “bumpy sphere” is contained inside a sphere of equation $\rho = a + b$. Find the values of θ and ϕ at which the two surfaces intersect.
 - (b) Use a CAS to graph the surface for $a = 14$, $b = 2$, $m = 4$, and $n = 6$ along with sphere $\rho = a + b$.
 - (c) Find an equation of the intersection curve of the surface at b. with the cone $\phi = \frac{\pi}{12}$. Graph the intersection curve in the plane of intersection.